1 Introduction

To be completed.

2 Materials and methods

2.1 *Phaeocystis globosa* culture condition

The *Phaeocystis globosa* strain used in this experiment was obtained from the Key Laboratory of Marine Environment and Ecology, College of Environmental Science and Engineering, Ocean University of China. Seawater used for cultivation was collected from coastal areas in Qingdao (salinity around 35), then filtered through 4.5 μm filter paper, autoclaved with 121°C and 100 kPa for 20 min using autoclave. After seawater cool to room temperature, f/2 nutrient medium were added according to Table [*]. In CNS, no nitrate was added in f/2; in CPS, no phosphate was added in f/2.

2.2 Experiment design

The experiment consists of three main treatments: CC (control group), receiving full f/2 medium; CNS (nitrate starvation), receiving f/2 medium without nitrate; CPS (phosphate starvation), receiving f/2 medium without phosphate. Each treatment was duplicated once (CC-1, CC-2; CNS-1, CNS-2; CPS-1, CPS-2). The duration of the experiment is 4 days. Samples were collected daily for cell counting and chlorophyll fluorescence measurement. On day 0 and day 4, nitrate and phosphate concentration measurements were carried out. On day 4, samples were collected and prepared for transcriptome analysis.

2.3 *P. globosa* cell counting

1 mL of sample was collected daily at a set time and fixed with Lugol's iodine. After thorough mixing, sample was counted under microscope with 100 μL sample volume. Noticed that *P. globosa* has two life forms of colonies and free living cells, numbers of colonial cells (Y) were calculated using the empirical model from (*), where diameter (X, μm) of colony was measured under microscope:

$$Y = 0.0079X^{2.1271} \tag{1}$$

The total cell density (cells/mL) was calculated as the sum of numbers of colonial cell density and solitary cell density.

2.4 Measurement of chlorophyll fluorescence

2 mL of sample was collected daily at a set time and dark-adapted for 15 min. The measuring light was switched on to record the minimum fluorescence (F_0) of PSII after signal stabilization, followed by a saturating pulse to measure the maximum fluorescence (F_m) and calculate the maximum quantum yield (F_v/F_m). Rapid Light Curve (RLC) were constructed under various photosynthetically active radiation (PAR) gradients and non-linearly fitted with the EP model to obtain the light utilization efficiency (α) and maximum electron transport rate ($\mathrm{ETR_{max}}$), then the Light Adaption Parameter (I_k) was further obtained.

2.5 Measurement of nitrate and phosphate concentration

Nitrate: Nitrate in the culture is determined under *Environmental Protection Industry Standard of the People's Republic of China* (HJ/T 346-2007) by ultraviolet spectrophotometry. Samples were filtered through 0.45 μm mixed cellulose ester membranes. The method is based on the absorption of nitrate ions at 220 nm and the second measurement at 275 nm was performed to correct the nitrate value. The corrected absorbance was calculated as:

$$A_{\mathrm{corr}} = A_{220} - 2A_{275} \tag{2}$$

where A_{220} and A_{275} are the absorbance at 220 nm and 275 nm, respectively. A calibration curve was obtained using a nitrate nitrogen standard stock solution within the range of 0.25–2.00 mg/L nitrate nitrogen. Then, 1.0 mL of HCl solution and 0.1 mL of sulfamic acid solution were added to the sample. Absorbance was measured using a 10 mm quartz cuvette with fresh deionized water (passed through the resin column and acidified with HCl) as the reference. The nitrate nitrogen concentration was calculated from the calibration curve. If dilution was applied, the result was multiplied by the dilution factor.

Phosphate: Phosphate in the culture is determined under *Specifications for Oceanographic Survey of the People's Republic of China* (GB/T 12763.4—2007) by ultraviolet spectrophotometry. Samples were filtered through 0.45 μm mixed cellulose ester membranes. Then, 25 mL of sample was transferred into 50 mL reaction bottle, along with 2.0 mL of sulfuric acid–ammonium molybdate–potassium antimonyl tartrate mixed reagent and 0.5 mL of ascorbic acid solution (54.0 g/L). The mixture was allowed to develop color for 10 min. Absorbance was measured at 882 nm using a 10 cm cuvette with distilled water as the reference. A calibration curve was constructed using $\mathrm{PO_4^{3-}}$-P standard solutions at 0, 0.40, 0.80, 1.60,

3.20, and 4.80 μmol/L. The concentration of reactive phosphate was calculated as:

$$[\mathrm{PO_4^{3-}\text{-}P}] = \frac{(A_w - A_b) - a}{b}, \tag{3}$$

where A_w is the mean absorbance of the sample, A_b is the blank absorbance, and a and b are the intercept and slope of the calibration curve, respectively.

2.6 RNA extraction and preparation

To be completed.

2.7 Transcriptome data analysis

To be completed.

3 Results

3.1 Growth and total cell density

The total cell density of *P. globosa* increase in all treatments during four days. CC and CNS reach their max cell density in day 3, while CPS continue its increase in day 4. The highest cell density was observed in CC day 3, of 1.38×10^6 cells/mL, followed by CNS of 1.14×10^6 cells/mL. In day 4, both CC and CNS had decrease in cell density, while CPS continued its increase to 6.37×10^5 cells/mL.

3.2 Photosynthetic performance

F_v/F_m, the maximum quantum yield of Photosystem II was measured daily to evaluate the photosynthetic performance. In CC, F_v/F_m decreased slightly from 0.7 to 0.64 from day 0 to day 4. In CNS, F_v/F_m declined slightly from 0.62 to 0.55 from day 0 to day 4. In CPS, F_v/F_m was relatively constant, remaining around 0.62–0.64 throughout the experiment.
α, the light utilization efficiency, remained relatively stable in CC and CPS throughout the experiment (Fig.). In CC, α steadily increased from 0.0208 in day 0 to 0.0228 in day 4. In CPS, α fluctuated between 0.207 on day 1 and 0.234 on day 2, and dropped to 0.213, nearly the same level on day 0. In CNS, α was 0.212 on day 0, remained around 0.212–0.219 until day 3, and then decreased to 0.195 on day 4. On day 4, α was highest in CC (0.213), followed by CPS (0.213) and CNS (0.195).

I_k, the light adaptation parameter, fluctuated considerably in all treatments, where CC and CNS showed a rising trend. In CC, I_k ranged from 1017.45 ± 266.10 Lux on day 1 to 1850.20 ± 1034.39 Lux on day 2, and was 1845.75 ± 737.16 Lux on day 4. In CNS, I_k was 813.25 ± 7.57 Lux on day 0, increased sharply to 2729.10 ± 1398.48 Lux on day 1, decreased to 1324.60 ± 98.29 Lux on day 2, and then increased again to 2318.85 ± 1455.29 Lux on day 4. In CPS, I_k varied between 1223.50 ± 126.29 Lux on day 3 and 1919.00 ± 699.05 Lux on day 2, and was 1482.35 ± 165.82 Lux on day 4. On day 4, I_k was highest in CNS (2318.85 ± 1455.29 Lux), followed by CC (1845.75 ± 737.16 Lux) and CPS (1482.35 ± 165.82 Lux).

3.3 Nitrate and phosphate concentration

Nitrate and phosphate concentrations in the culture media were measured on day 0 and day 4 (Fig.). In all treatments, nitrate concentration decreased around
In CC, nitrate decreased slightly from 766.47 μmol/L on day 0 to 720.32 μmol/L on day 4. In CPS, nitrate remained high, decreasing from 726.38 μmol/L on day 0 to 702.92 μmol/L on day 4. In CNS, nitrate was already low on day 0 (52.38 μmol/L) and further declined to 23.78 μmol/L on day 4.
For phosphate, CC and CNS maintained relatively high concentrations throughout the experiment. In CC, phosphate decreased from 32.33 μmol/L on day 0 to 24.29 μmol/L on day 4. In CNS, phosphate decreased slightly from 31.55 μmol/L on day 0 to 27.19 μmol/L on day 4. In CPS, phosphate was low on day 0 (2.35 μmol/L) and further decreased to 0.70 μmol/L on day 4.
These results confirm the successful establishment of nitrogen starvation and phosphorus starvation conditions, respectively.

3.4

To be completed.

4 Discussion

To be completed.